

Mathematics in Zombieland

Spreadsheet-Aligned Formula Guide

SUDZ state variables • kbamzr rate variables • exact iterative calculations

Purpose and source

This guide rewrites the mathematical model exactly as it is implemented in **miz-model.xlsx** (the same file as “1 year model mathematics in zombieland v2.xlsx”). The workbook advances the population one numbered iteration at a time; it does not directly evaluate continuous-time derivative equations. Every population result is also prevented from becoming negative by Excel's **IF(...<0,0,...)** wrapper.

Notation

n	current iteration; $n+1$ is the next iteration	columns B, C, D, ...
h	interaction/time-step scaling constant	\$C\$3; default 0.000002
S_n	susceptible people at iteration n	row 4
U_n	undead/infected-but-not-yet-zombie people	row 5
D_n	cumulative dead at iteration n	row 6
Z_n	zombies at iteration n	row 7
k_n	fraction/probability of zombie-killing encounters	row 11
b_n	birth rate applied to susceptible people	row 12
a_n	fraction/probability of encounters that infect a susceptible person	row 13
m_n	non-zombie mortality rate applied to susceptible people	row 14
z_n	fraction/probability of undead eligible to complete zombification	row 15
r_n	fraction of undead converted to zombies per iteration	row 16

SUDZ names the four population states. **kbamzr** names the six rates in the order used by the workbook: k , b , a , m , z , r .

Exact SUDZ update formulas

Define **clamp**(x) = $\max(0, x)$. Then the workbook calculates:

Susceptible: $S_{n+1} = \text{clamp}[S_n + h(b_n S_n - m_0 S_n - a_0 S_n Z_n)]$

Undead: $U_{n+1} = \text{clamp}[U_n + h(a_n S_n Z_n - z_n U_n) - r_n U_n]$

Dead: $D_{n+1} = \text{clamp}[D_n + h(k_n S_n Z_n + m_n S_n)]$

Zombies: $Z_{n+1} = \text{clamp}[Z_n + z_n r_n U_n - h k_n S_n Z_n]$

Workbook-specific detail: the susceptible formula fixes a and m to baseline cells B13 and B14

(shown as a_0 and m_0), even after later columns change a_n or m_n . The undead, dead, and zombie

formulas use rates from the current iteration. This guide preserves that behavior for spreadsheet fidelity.

What each update adds and subtracts

- **Susceptible:** begin with S_n ; add scaled births $h b_n S_n$; subtract scaled ordinary deaths $h m_0 S_n$; subtract scaled infections $h a_0 S_n Z_n$.
- **Undead:** begin with U_n ; add scaled infections $h a_n S_n Z_n$; subtract scaled zombification eligibility $h z_n U_n$; then subtract the per-iteration converted share $r_n U_n$.
- **Dead:** begin with D_n ; add scaled zombies killed $h k_n S_n Z_n$ and scaled ordinary deaths $h m_n S_n$. This row is cumulative.
- **Zombies:** begin with Z_n ; add undead who complete conversion $z_n r_n U_n$; subtract scaled zombies killed $h k_n S_n Z_n$.

Excel cell form for one step (B → C)

These are simplified but cell-faithful versions of C4:C7:

C4 = MAX(0, B4 + \$C\$3 × (B12×B4 – \$B\$14×B4 – \$B\$13×B4×B7))

C5 = MAX(0, B5 + \$C\$3 × (B13×B4×B7 – B15×B5) – B16×B5)

C6 = MAX(0, B6 + \$C\$3 × (B11×B4×B7 + B14×B4))

C7 = MAX(0, B7 + B15×B5×B16 – \$C\$3×B11×B4×B7)

Interactions and cumulative interactions

- Interactions at iteration n : $I_n = h S_n Z_n$
- Cumulative interactions: $T_1 = I_1$ and $T_n = T_{n-1} + I_n$

How kbamzr connects to SUDZ

k	moves killed zombies out of Z and adds the same scaled quantity to D	D, Z
b	adds births to S	S
a	removes infected people from S and adds infections to U; S uses baseline a_0	S, U
m	removes ordinary deaths from S and adds them to D; S uses baseline m_0	S, D
z	participates in loss from U and gain to Z, but with different scaling in the workbook	U, Z
r	removes $r_n U_n$ from U and contributes $z_n r_n U_n$ to Z each iteration	U, Z

Corrections to the short formulas in the task guide

- “Change in Zombies = $zU - kSZ$ ” omits both h and r . The workbook update is $Z_{n+1} = \text{clamp}[Z_n + z_n r_n U_n - h k_n S_n Z_n]$.
- “Change in Undead = $aSZ - zU$ ” omits h , r , the prior U value, and the workbook’s placement of r outside h . Use the full U update above.
- “Change in Dead = $kSZ + mS$ ” captures the two flows but omits h , the prior cumulative D value, and the zero clamp.
- “Change in Susceptible = $bS - mS - aSZ$ ” captures the three flows but omits h , the prior S value, the clamp, and the workbook’s fixed baseline references for a and m .
- The task guide states that the current birth rate is 0.03. The workbook’s **No Action** baseline is **0.003**; 0.03 begins in Phase 4. Use the value in the relevant workbook column.

Reading the model correctly

- Each new column uses the previous column’s S , U , D , and Z values and the previous column’s rate settings.
- The IF wrapper turns any negative proposed population into zero; the workbook does not permit negative SUDZ values.
- The factor h scales births, deaths, encounters, infection, and the zU subtraction in U , but it does not scale rU in U or zrU in Z .
- The spreadsheet is not perfectly mass-conserving. Explain results as outputs of this workbook rather than assuming every person removed from one row appears identically in another.

Prepared from Sheet2, rows 4–16, in the v2/published workbook. Originals were not modified.